

Delius Schoeman

SOFTWARE DEVELOPER | FULL-STACK WEB DEVELOPER | AI & MACHINE LEARNING

PROFESSIONAL SUMMARY

Information Technology graduate currently completing an Honours degree in Information Technology at North-West University, expected in November 2026. Passionate about software development, full-stack web development, artificial intelligence and machine learning. Experienced in developing software solutions through academic and personal projects, including full-stack web applications, third-party API integrations and machine learning research. Strong problem-solving and analytical skills with a commitment to developing reliable, scalable and user-focused software solutions.

EDUCATION

Bachelor of Information Technology Honours, North-West University

Expected Completion: November 2026

Research Topic: Multimodal Machine Learning Framework for the Early Detection and Prediction of Foot-and-Mouth Disease Outbreaks in South Africa.

Bachelor of Information Technology, North-West University

Graduated: 2025

ACHIEVEMENTS

- Top Performing Student – Achieved the highest final mark (93%) in an Information Technology module (CMPG 122 – UI Programming) at North-West University.
- Highest Achievement in Information Technology – Grade 12 & 11.

PROJECTS

Multimodal Machine Learning Framework for Foot-and-Mouth Disease Detection (Honours Research)

- Python, PyTorch, OpenCV, Machine Learning.
- Developing a multimodal ML framework integrating image, textual and structured environmental data.

Campus Marketplace Web Application

- Firebase, React, Node.js, CSS, Paystack API.
- Built a full-stack marketplace with authentication, payments and a relational database.
- Group Project

Marketplace Website

- PHP, MySQL, HTML, CSS, JavaScript, Google Maps API, Paystack API, The Courier Guy API.
- Developed a full-stack marketplace with authentication, payment processing, location services and delivery integration.
- Solo project

Personal Portfolio Website

- HTML, CSS, JavaScript.
- Responsive portfolio showcasing technical skills and projects.
- Live: ystervarkies.github.io/Delius-Schoeman-Portfolio

Morabaraba - AI Game Agent

- Python, Tkinter, Reinforcement Learning (Q-Learning).
- Built a self-learning Morabaraba game-playing agent for a university AI module, using tabular Q-learning with epsilon-greedy exploration and self-play training, including formal TEP (Task, Environment, Performance) framework design and experimental evaluation.

DETAILS

📞 079 218 3025

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LINKS

[Portfolio Website](#)

[GitHub](#)

[LinkedIn](#)

TECHNICAL SKILLS

- Python
- C#
- C++
- PHP
- JavaScript
- SQL
- HTML5
- CSS3
- React
- Next.js
- ASP.NET
- Bootstrap
- MySQL
- Firebase
- Machine Learning
- Computer Vision
- PyTorch
- Tensorflow
- OpenCV
- API
- Git
- GitHub
- Visual Studio
- Visual Studio Code
- XAMPP
- MATLAB

- Developed a Python/Tkinter GUI with JSON-based model export, and produced a technical report analysing agent performance.

EXPERIENCE

IT Vacation Student, Iclix (Pty) Ltd - July 2024

- Assisted with software development and IT-related tasks.
- Worked alongside experienced developers and gained exposure to software development workflows.

Construction Labourer (Vacation Employment) - recurring seasonal work: December 2022, December 2023, December 2024, December 2025, July 2026

- Worked during July and December university vacations.
- Developed teamwork, communication, reliability and time management skills.

Structured Programming (CMPG 121) Student Assistant – July 2026 until October 2026

- Assisted students with fundamental programming concepts in CMPG121 (Structured Programming), providing guidance during practical sessions, answering programming-related questions, and supporting the development of problem-solving and coding skills using C++.

Apps and Advanced User Interface Programming (CMPG212) Student Assistant – February 2026 until May 2026

- Assisted students with practical programming concepts in CMPG212 (Apps and Advanced User Interface Programming), providing guidance during practical sessions, answering technical questions, and supporting the development of C# and .NET applications.

User Interface Programming (CMPG122) Student Assistant – May 2024 until October 2024

- Assisted students with practical programming concepts in CMPG122 (Apps and Advanced User Interface Programming), providing guidance during practical sessions, answering technical questions, and supporting the development of C# and .NET applications.

PROFESSIONAL SKILLS

Software Development • Full-Stack Web Development • Machine Learning • Problem Solving • Analytical Thinking • Communication • Teamwork • Adaptability • Critical Thinking • Time Management

REFERENCES

Professor Dimitri Keykaan

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Professor Absalom Ezugwu

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